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WHO Ambient Air quality database

Air Quality database Update 2026 (Version Eight)
Released on 30 June 2026 – 8th edition (V8.0)

The WHO Ambient Air Quality Database compiles ground-level measurements of annual mean concentrations for nitrogen dioxide (NO_2), and particulate matter with a diameter equal to or smaller than 10 µm (PM_{10}) and 2.5 µm ($PM_{2.5}$). These measurements aim to represent an average for the city or town, rather than for individual monitoring stations.

Both groups of pollutants originate mainly from human activities related to fossil fuel combustion. To ensure the data accurately reflects population exposure, the database primarily utilizes urban measurements, encompassing urban background, residential, commercial, and mixed areas, as well as rural and industrial zones close to urban settlements.

- **Frequency:** Updated regularly every 2–3 years since 2011.
- **Impact:** This data serves as the primary input to derive Sustainable Development Goal (SDG) Indicator 11.6.2 (Air quality in cities), for which WHO is the custodian agency.
- **Primary Sources:** Official national and subnational datasets provided by countries upon request, official reports, and government websites. For European Union members, data is sourced from the European Environment Agency (EEA).

This version of the database (V8.0, released in June 2026) currently hosts data from 8 023 human settlements in 127 WHO Member States, WHO Associate Members or WHO Territory or Areas, for the period 2010-2024.

Limitations

While the database provides critical insights, data comparability between different countries is limited due to the following factors:

- *Coverage and Accessibility Gaps:* The database is not comprehensive. It captures only a small fraction of cities in some countries, and certain existing data could not be accessed due to language barriers or limited availability.
- *Methodological and Quality Differences:* There is large heterogeneity in measurement methods, techniques, and data quality across regions. Additionally, some data points may lack sufficient documentation to fully verify their compliance with inclusion criteria.
- *Spatial and Temporal Variations:* Measurements are localized (city averages are based on ground stations) and temporal coverage varies. Data spanning only part of the year may significantly deviate from the true annual mean due to seasonal variability.

How this data is used

The WHO Ambient Air Quality Database serves these critical functions:

✅ *Advancing Air Pollution Exposure Modeling*: It provides foundational input for complex models—combining ground measurements with topographic, population, satellite data, and chemical transport models—to accurately estimate global air pollution exposure. This modeling framework ensures homogeneous and standardized exposure estimates across all countries, regardless of local monitoring capacity.

✅ *Driving Advocacy and Transparency*: It raises awareness about the vital importance of open-access, high-quality data, while actively advocating for the expansion and improvement of air monitoring networks worldwide.

✅ *Quantifying the Global Health Burden*: It acts as a cornerstone for calculating the mortality and morbidity burdens attributed to air pollution. WHO estimates that outdoor air pollution caused 4.3 million premature deaths in 2021 due to stroke, ischaemic heart disease, lung cancer, chronic obstructive pulmonary disease and acute lower respiratory infections.

How this data is not used

To ensure the database is used accurately and responsibly, it should not be used to:

❌ *Rank the most polluted cities:* The database is not designed to create city rankings, as monitoring capacities and data completeness vary significantly between locations.

❌ *Draw direct comparisons between countries*: Differences among monitoring networks, and measurement methodologies make direct cross-country comparisons misleading.

Related pages

- Air quality database 2024 (V6)
- Air quality database 2022 (V5)
- Air quality database 2018 (V4)
- Air quality database 2016 (V3)
- Air quality database 2014 (V2)
- Air quality database 2011 (V1)
- Ambient air pollution data
- Submitting official data

Resources

The following documents are relevant to the current version of the WHO Ambient Air Quality database (Version 2024 V6.1).

Excel Version V8.0 - WHO Ambient Air Quality Database (Update June 2026)

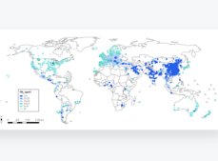
The suggested citation for the database is: World Health Organization (2026). WHO Ambient Air Quality Database (Version 8.0.) Geneva, World Health Organization, 2026.

Interactive web application V8.0 (Update June 2026)

The App includes built-in WHO data, originating from the WHO Ambient Air Quality Database V8.0 (Update June 2026). The suggested citation for the app is: World Health Organization (2026). WHO Ambient Air Quality database: interactive tool (Version 8.0). Geneva: World Health Organization, 2026.

Related article

WHO air quality database: relevance, history and future developments
This article published in the WHO bulletin in 2023, presents an overview of close to 50 years of WHO's activities in gathering air quality data and their use in global assessments and efforts to reduce health risks caused by air pollution.



WHO air quality database: relevance, history and future developments

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Report (Version 2022 V5.0)

1 June 2023
WHO ambient air quality database, 2022 update: status report

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Additional documents

The following documents are relevant to the previous (Version 2022 V5.0) and current version of the WHO Ambient Air Quality database.

Methods used for compiling the database

Frequently asked questions about the database

Related indicator

Knowing the level of air pollution all over the world is critical to enable public action. Such robust data are essential for countries and the global community to establish a baseline, track progress, draw comparison, and advocate for actions. WHO is a custodian agency for the Sustainable Development Goals (SDG) indicator 11.6.2 Annual mean levels of fine particulate matter (e.g. PM2.5 and PM10) in cities (population weighted), as displayed below. This indicator aims to inform progress on the target 11.6 set by the United Nations as “By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management:, as part of the broader goal 11 “Make cities and human settlements inclusive, safe, resilient and sustainable”.

Because ground monitors are not available everywhere, WHO uses modelled estimates, which rely, among other input data, on the information available in the WHO Ambient Air Quality Database. More information about the modelled estimates from DIMAQ can be found here.

Visualisations | Data | Metadata | Related indicators

Indicator 11.6.2: Annual mean levels of fine particulate matter (e.g. PM2.5 and PM10) in cities (population weighted) SDG Indicator 11.6.2 Concentrations of fine particulate matter (PM2.5)

Related factsheet

24 October 2024
Ambient (outdoor) air pollution

Related health topic

Physical environment
Air pollution

Related tools

- Air quality standards tool
- AirQ+ tool

Related teams

- Air quality, energy and health
- Environment, Climate Change, One Health and Migration

Additional Resources

- WMO Global Atmospheric Watch
- UNEP Actions on Air Quality
- UNEP Regulating Air Quality
- CCAC: Air Pollution in Asia and the Pacific: Science-based solutions
- UNECE Guidance Document on Emission Control Techniques for Mobile Sources Under the Convention on Long-range Transboundary Air Pollution
- European Environment Agency - Air Quality in Europe

Contact information for general inquiries: aqh_who@who.int

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